

THE CHOICE LINE

AI is present. People retain the choice.

Others test whether AI can be withdrawn. Choice Line tests whether declining AI still allows the same task to be completed without penalty.



1—3—2—12—6—1

No penalty for declining AI

Service Twin turns choice into a deliverable.

01

CHOICE FIRST

See both routes before collection

02

SAME OUTCOME

One task and one accepted outcome

03

COMPLETABLE

Offline, staffed, and phone-free completion

04

TAKEOVER

Immediate human takeover on failure

05

APPEALABLE

Receipt, owner, and response time

06

MEASURED

Proposed gate: CEI $\geq$ 90 and each item $\geq$ 80

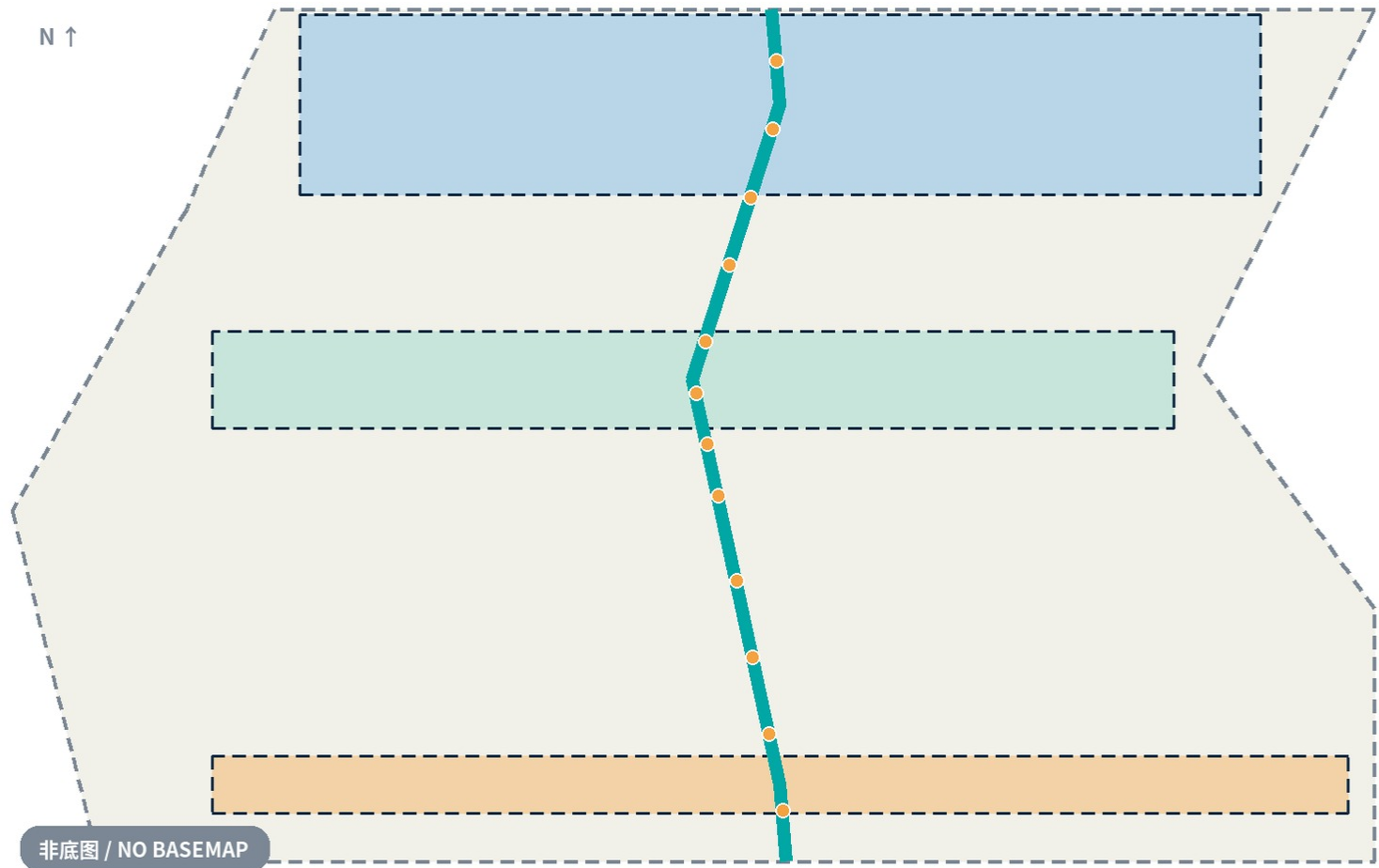
NO PENALTY FOR DECLINING AI

SERVICE TWIN / SIX ACCEPTANCE SAFEGUARDS

PLATE 03

One evidence chain, three scales

Overview map • provisional scope



Three scopes and evidence chain

43.6 km²

Research • announcement text

≈ 11.4 km²

Overall design • announced value

368.4 ha

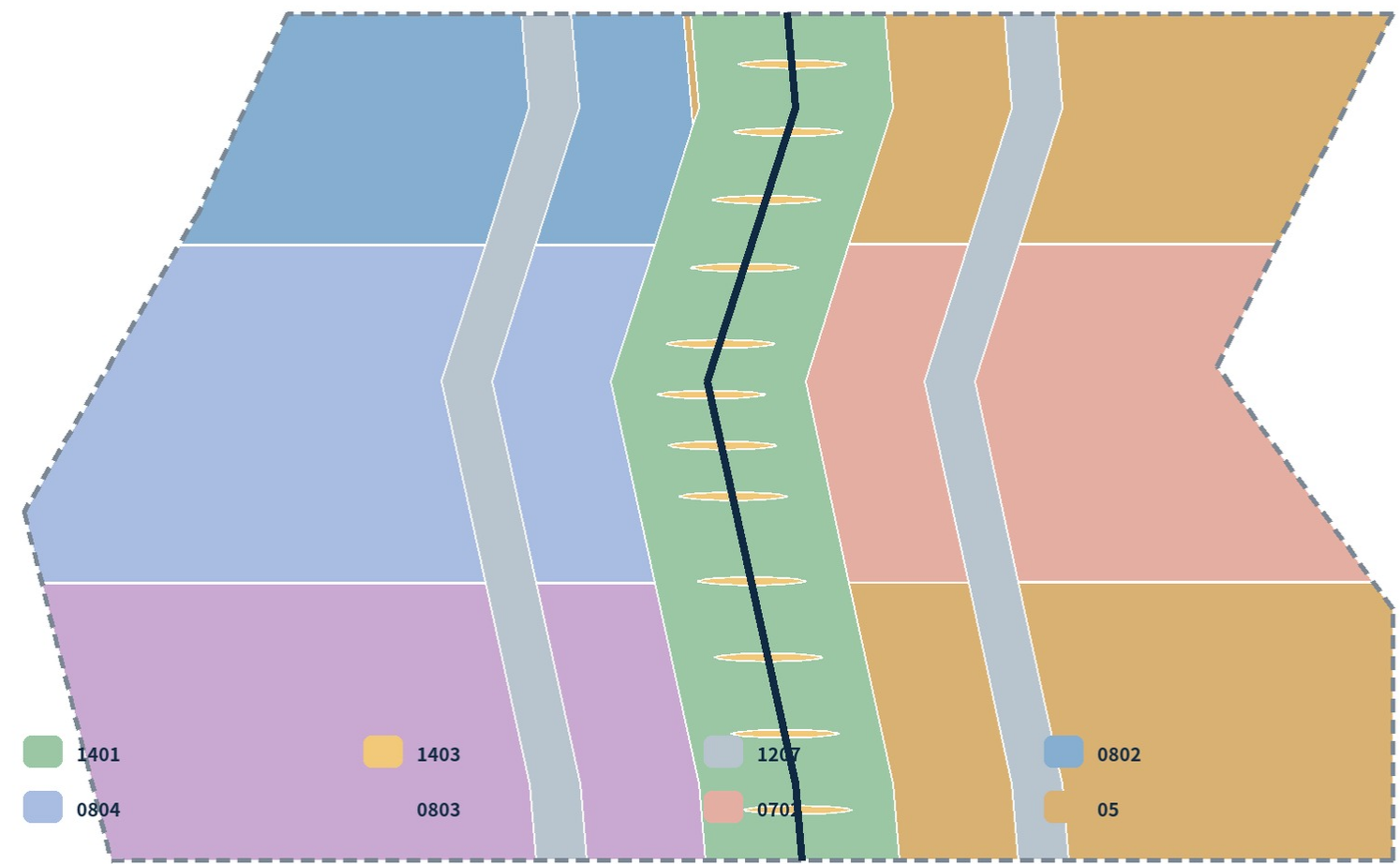
Three key areas • announced sum

- 1 Announcement / taskbook
- 2 Repository provisional geometry
- 3 Nine GeoJSON layers
- 4 EPSG:4548 metrics
- 5 Scenarios and risks
- 6 Bilingual figures / PDF

All alignments await survey and rights checks

The 1-3-2-12-6-1 framework

Land-use partition • topologically closed



Spatial mnemonic

1-3-2-12-6-1

- 1 Ground Truth Walk
- 3 choice labs
- 2 collaborative wings
- 12 Service Twin nodes
- 6 choice safeguards
- 1 CEI disclosure system

Jing-Zhang culture

urban AI life

AI-integrated innovation

Three zones form an evidence pipeline

NORTH • ZHONGZHIYUAN

FAIL-SAFE GARDEN

- 1 Robot safe stop and human handover**
Trial operator and on-site safety marshal
- 2 Model red-team and failure drill**
Independent assurance lead and service owner
- 3 Camera-free edge perception benchmark**
Data steward and facilities operator
- 4 Enterprise Agent interoperability and handover**
Enterprise service owner, interface engineer, and human handover...

Output: reproducible failure + takeover

PROVISIONAL RECTANGLE

MIDDLE • AI ORIGIN

OPEN LEARNING COMMONS

- 5 Account-free learning and research explanation**
Public-learning curator and duty explainer
- 6 Health-service navigation without diagnosis**
Health-navigation owner and qualified service worker
- 7 Legal and public-rule explanation without adjudication**
Public-service owner, rule editor, and independent reviewer
- 8 Accessible service co-test**
Accessibility lead, community representative, and facilities owner

Output: understandable rules + public review

PROVISIONAL RECTANGLE

SOUTH • DAZHONGSI

CIVIC CHOICE HUB

- 9 Station-park transfer navigation**
Mobility-service owner and on-site transfer steward
- 10 Bilingual Jing-Zhang cultural guide**
Cultural curator, history editor, and on-site interpreter
- 11 Anonymous and non-personalized retail**
Retail operator, consumer-rights contact, and shop worker
- 12 Face-free event flow**
Event owner, facilities maintainer, data steward, and safety marshal

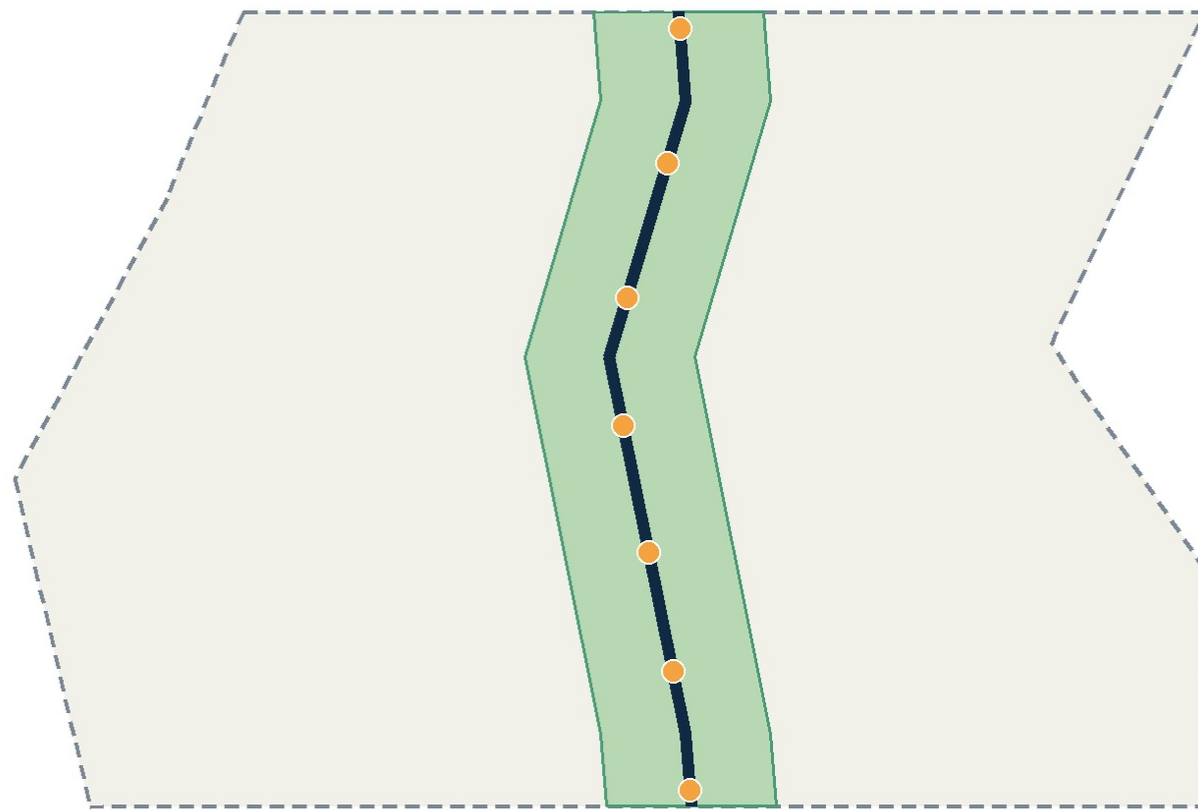
Output: real tasks + route gap

PROVISIONAL RECTANGLE

Discover failure → public review → real-service acceptance → anonymized feedback

One walk, equally visible twin interfaces

Slow mobility + blue-green public space



9.72 km

概念步道 / conceptual walk

7

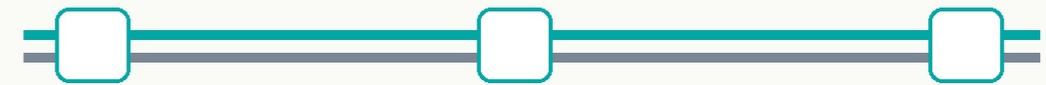
缝合门户 / stitching ports

15.29%

提案绿脊 / proposal green spine

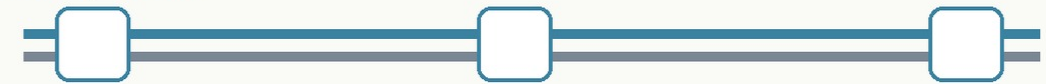
Three standard sections

Sensed / camera-free



notice before sensing + equal bypass

Smart / offline



screen, print, and tactile at equal prominence

Automatic / human

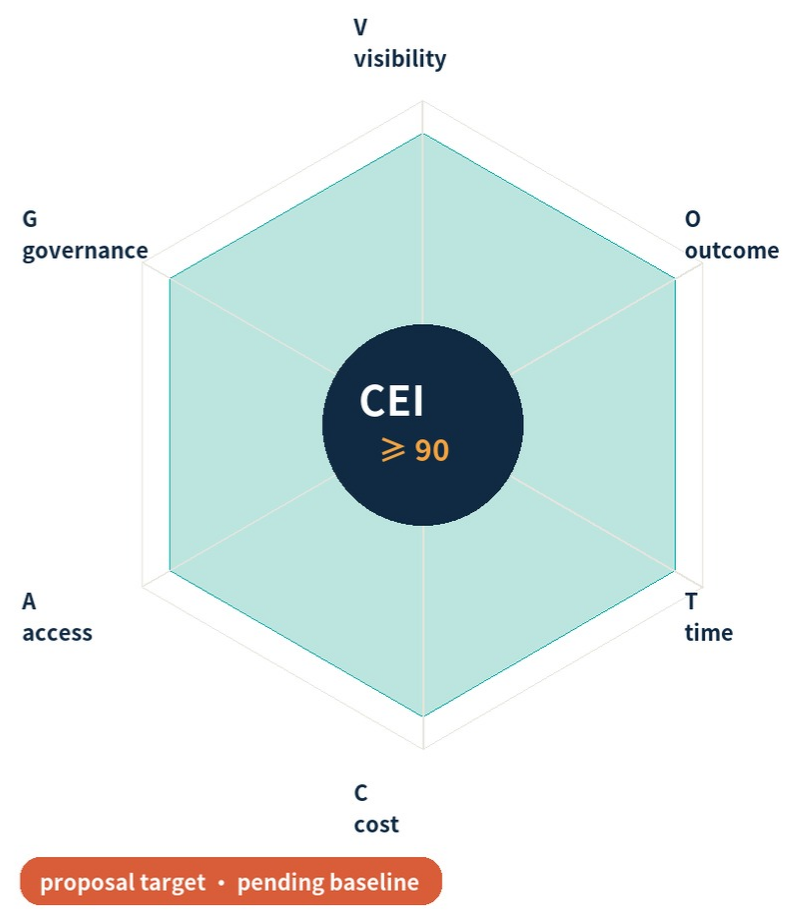


one case number, same price and hours

The task remains completable during network loss

Choice equivalence must be measured

CEI • Choice Equivalence Index



Core metrics + evidence closure

11.413 km²
provisional site area

15.29%
proposal green ratio

0.89%
node public space

12 / 7 / 4
nodes / ports / landmarks

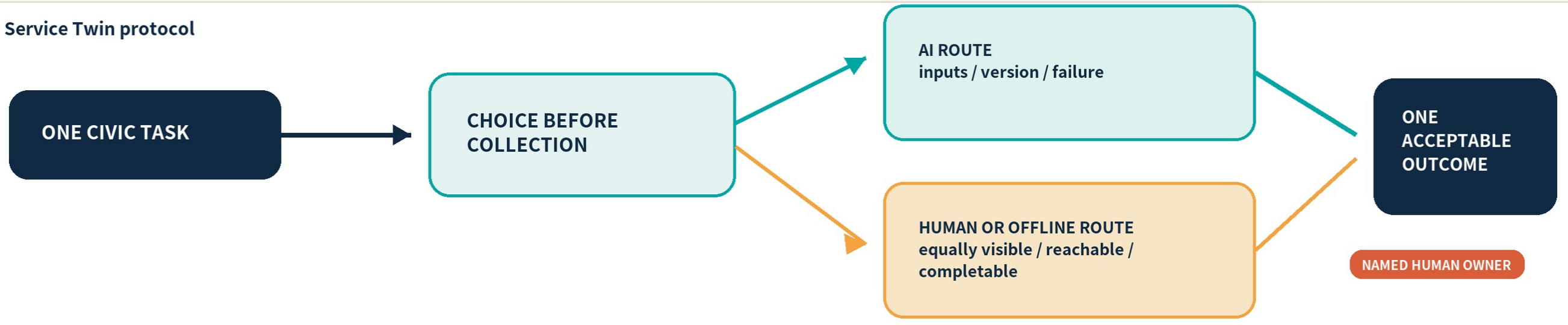


unknown is not zero: statutory FAR remains unknown

- Any component < 80: do not scale
- Weakest persona fails: no average override
- No funded human shift: no Service Twin
- Official polygon arrives: recalculate the package

One task, two completable routes

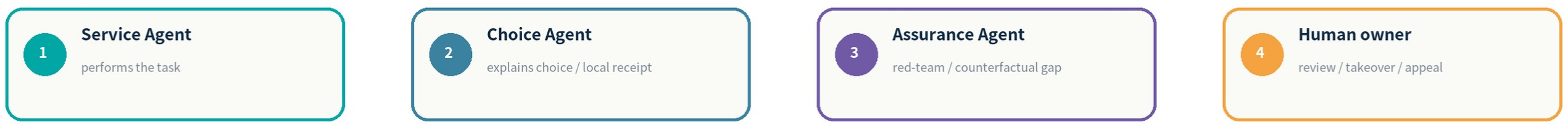
Service Twin protocol



$$CEI = 100 \times \text{mean}(V,O,T,C,A,G)$$

Any component < 80 or CEI < 90 → pause, remedy, transfer, or retire

Four-party loop: AI innovation does not automate accountability



Twelve scenarios that can stop

S01

Robot safe stop and human handover

DATA

Process sensors on device without uploading raw imagery; propose a 30-day minimal incident log subject to...

TWIN

A human courier completes the same task on the same accessible route with fee and hours displayed together

STOP

Stop immediately and isolate equipment after boundary breach, loss of link, or unrecognized works or obstacle

INDUSTRY ASSURANCE

robot-handover

S02

Model red-team and failure drill

DATA

Use no real personal data; de-identify logs and delete them under the project test plan

TWIN

Independent experts apply the same public checklist and publish findings beside model-based assurance

STOP

Pause after test-set contamination, irreproducible results, or escape of high-risk output

INDUSTRY ASSURANCE

model-red-team

S03

Camera-free edge perception benchmark

DATA

Collect no face, voiceprint, or device identifier and publish aggregate counts only

TWIN

Trained observers record the same non-identifying statistics during matched periods

STOP

Stop publication after identity inference, unsafe blind spots, or undersized aggregate samples

INDUSTRY ASSURANCE

camera-free-perception

S04

Enterprise Agent interoperability and handover

DATA

Use synthetic or cleared data only; tokens are short-lived and immediately revocable

TWIN

A human specialist completes the task using the same case number and exported material

STOP

Stop the pilot when export omits fields, permissions cannot revoke, or a person cannot restore service

INDUSTRY ASSURANCE

agent-interoperability

S05

Account-free learning and research explanation

DATA

Erase on-device preference at session end and build no learning profile

TWIN

A staffed explanation, print reader, and device-free labels deliver the same learning goal

STOP

Withdraw content after missing sources, claims beyond evidence, or absent accessible format

PUBLIC SERVICE

account-free-learning

S06

Health-service navigation without diagnosis

DATA

Store no symptoms and infer no disease; send appointment essentials directly to the accountable...

TWIN

Phone, print directory, and trained staff complete the same search and appointment help

STOP

Transfer immediately after diagnostic language, stale provider data, or urgent risk

PUBLIC SERVICE

health-navigation

S07

Legal and public-rule explanation without adjudication

DATA

Do not request unnecessary identifiers and erase temporary context when the session ends

TWIN

An in-person desk, phone line, and print checklist explain the same case under one identifier

STOP

Stop automated response after uncertain version, conflicting rules, or a rights determination

PUBLIC SERVICE

rules-explainer

S08

Accessible service co-test

DATA

Collect no biometric data; de-identify feedback and permit withdrawal

TWIN

Physical wayfinding, staffed accompaniment, and a print checklist complete the same task

STOP

Pause after a critical route break, non-withdrawable feedback, or inconsistent safety information

PUBLIC SERVICE

accessible-co-test

S09

Station-park transfer navigation

DATA

No persistent location tracking; erase on-device location at session end

TWIN

Physical signs, a paper route map, and staffed help complete the same arrival task

STOP

Stop recommendations after lift failure, stale information, or an inaccessible route break

PUBLIC SERVICE

station-park-transfer

S10

Bilingual Jing-Zhang cultural guide

DATA

Build no visitor profile and erase optional session preferences at the end

TWIN

Physical labels, paper map, and a human guide independently deliver the full narrative

STOP

Withdraw after unknown provenance, distorted translation, or describing a proposal as implemented

PUBLIC SERVICE

jingzhang-cultural-guide

S11

Anonymous and non-personalized retail

DATA

Collect no face, device identifier, or cross-store history and erase the session at exit

TWIN

A staff member and paper category index complete shopping at the same price and hours

STOP

Disable after price disparity, sensitive inference, manipulative interface, or failed session deletion

PUBLIC SERVICE

anonymous-retail

S12

Face-free event flow

DATA

Retain time-banded aggregate counts only and delete raw counts after the event under the published rule

TWIN

Human patrol, physical zone signs, and public address deliver the same safety operation

STOP

Disable after distorted counts, storm threshold, blocked accessible exit, or failed deletion

PUBLIC SERVICE

face-free-event-flow

The least-served persona gates expansion

Eight personas are acceptance tasks, not labels

- 1

P01 • Older adult or feature-phone user

Complete the service without an app, account, or QR code using large-print, phone, and in-person support.

Exclusion risk: Forced device purchase, longer queues, or denial because a digital credential is missing.
- 3

P03 • Parent, caregiver, or minor

Shared plain-language notice before collection, clear guardian boundaries, care space, and no profiling of minors.

Exclusion risk: Default consent, manipulative interfaces, retained child data, or stroller routes omitted by automated advice.
- 5

P05 • International visitor or developer

Bilingual rules, short-term account-free access, cross-cultural explanation, and clear participation and exit limits.

Exclusion risk: Chinese-only digital interfaces, inaccessible local accounts, or mistaking an exhibition for official approval.
- 7

P07 • SME developer or startup team

Affordable interoperability tests, compliance help, independent assurance, prototyping, and a defined retirement interface.

Exclusion risk: Single-vendor lock-in, unaffordable repeated assurance, or stranded data and services after a pilot.

- 2

P02 • Commuter with visual, hearing, or mobility disability

A continuous accessible route, tactile and audio information, captions, seating, and a camera-free option.

Exclusion risk: Broken station-park continuity, hidden alternatives, or assistive and staffed services unavailable together.
- 4

P04 • Maintenance, cleaning, or delivery worker

Safe loading, rest and charging points, clear machine takeover rights, isolation procedures, and humane schedules.

Exclusion risk: Blocked by robot trials, uncompensated takeover labor, or penalties from performance models without appeal.
- 6

P06 • University student or researcher

Open evaluation, quiet study, explainable research, night safety, exercise, and affordable daily services.

Exclusion risk: Innovation space reserved for corporate users, opaque results, or disconnected night routes and study facilities.
- 8

P08 • Human reviewer or public-service worker

Safe workstations, shift capacity limits, training, takeover authority, appeal logs, and protection from automation pressure.

Exclusion risk: Human fallback treated as limitless free labor while reviewers absorb model errors without stop authority or rest.

Weakest-persona gate

A ≠ average satisfaction

A compares the smaller and larger priority-persona success rates. If any group is punished by fallback, the node does not scale.



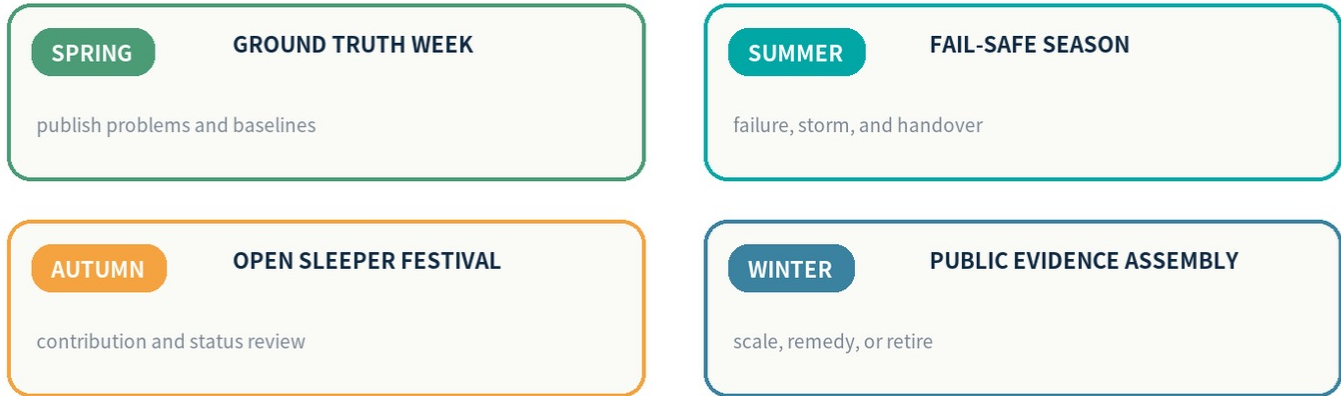
No component below 80

Evidence first, anchors second, scale last

Three stages: every expansion has a stop condition



Four-season operations

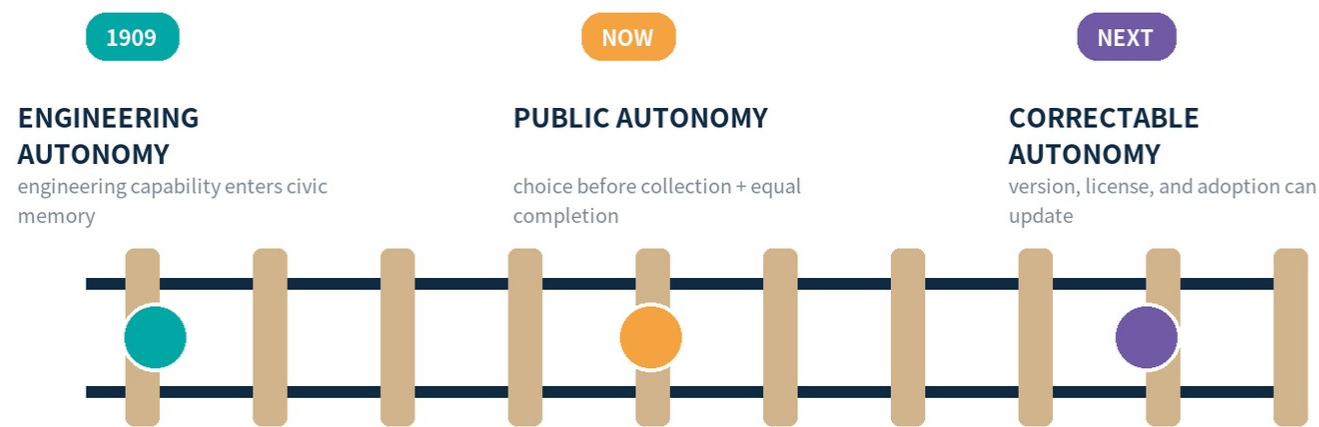


Responsibility and funding loop



From engineering autonomy to public autonomy

Centennial narrative: autonomy is not a one-off memorial



Others test whether AI can be withdrawn. Choice Line tests whether a person can finish the same task without being punished for declining AI.

不声称 ‘全球第一’ / NO WORLD-FIRST CLAIM

Four landmarks + open-sleeper state machine

- LM-01

Choice Mile Zero

Extend the engineering autonomy of 1909 into public autonomy to understand, choose, and keep receiving service in the AI era.
- LM-02

Human Review Hall

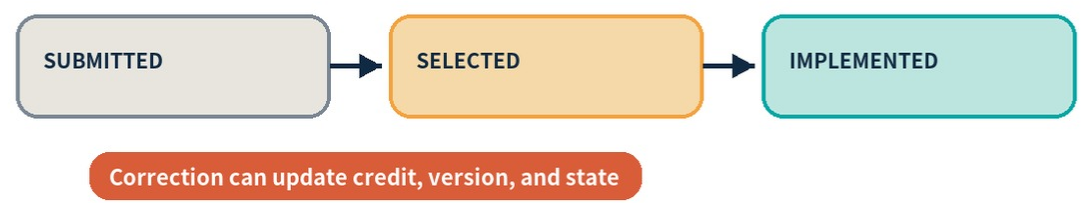
Turn model limits, appeals, takeover, and accountable ownership from a back-office process into visible civic space.
- LM-03

Fail-Safe Garden

Publicly demonstrate when a system should stop, who can stop it, and how essential urban service continues afterward.
- LM-04

Open Sleeper Honor Line

Record Agents, people, versions, licenses, and actual adoption status in an updateable and correctable form.



Eight global cases, no superficial copying

Each case separates fact, inference, and non-transferable boundary.

CASE 01

Amsterdam algorithm register and human appeal

Publish a decision card for every rights-affecting service and provide a human review and non-digital appeal route independent of the original automated result.

The mechanism relies on Dutch administrative law and identity infrastructure; registry disclosure is neither independent audit nor universal opt-out.

CASE 03

Singapore Global AI Assurance Sandbox

Use a deployer, site owner, and independent tester triad in Zhongzhiyuan, publishing risk hypotheses, failure thresholds, remediation, and retirement records.

The sandbox is neither a software runtime nor regulatory approval; technical testing does not replace process governance, legal review, or foundation-model safety.

CASE 05

MIT Kendall Square / InnovationHQ

Use a city-open ground floor and specialized upper-floor program stack in the Origin Community, funding operations alongside prototyping, IP, mentors, and capital services.

MIT's research density, brand, land, and capital network are not transferable, and the evidence does not show that buildings alone cause commercialization.

CASE 07

New York City statutory open-data governance

Assign a data steward for each participating district, campus, park, and operator, publishing classification, update cadence, field definitions, quality checks, and correction tickets.

Only legally publishable data is open; a portal cannot guarantee completeness or fitness and does not replace privacy review or sustained maintenance staff.

CASE 02

Norwegian data-protection regulatory sandbox

Create a compliance co-design room in the AI Origin Community where legal, technical, operational, and public representatives agree test plans and publish reusable exit reports.

Sandbox advice is not approval, certification, or a binding decision; participants retain compliance responsibility and the jurisdictional structure cannot be copied.

CASE 04

Estonian X-tee / X-Road exchange layer

Use source-retained data, service-specific authorization, revocable trust anchors, and verifiable call logs for cross-district cooperation rather than a single urban data lake.

X-Road is not identity, portal, or complete governance; endpoint authorization, data classification, and citizen-visible records still require separate design.

CASE 06

GOV.UK Assisted Digital and cross-channel service

Give phone, paper, in-person, and AI routes one service identifier; never hide the human channel, and feed offline failure evidence back into service improvement.

The guidance requires the channel mix users need, not every channel at all times, and disability access still needs its own research and formats.

CASE 08

Rotterdam Benthemplein water square

Design Dazhongsi event space for everyday use and stormwater storage, combining shade, seating, sediment cleaning, and programming in one maintenance brief.

Its storage capacity cannot be copied; runoff pollution, freeze-thaw, infiltration, child safety, and utility conditions require Beijing-specific calculation.

Twelve tasks, one delivery protocol

Data, twin route, stop rule, owner, and appeal are all mandatory.

S01

Robot safe stop and human handover

Complete low-speed delivery on a controlled park path and transfer safely to a person when risk triggers.

A human courier completes the same task on the same accessible route with fee and hours displayed together

S04

Enterprise Agent interoperability and handover

Prove a service can export, change provider, and transfer to a person after failure.

A human specialist completes the task using the same case number and exported material

S07

Legal and public-rule explanation without adjudication

Turn public rules into understandable service steps and identify the accountable authority.

An in-person desk, phone line, and print checklist explain the same case under one identifier

S10

Bilingual Jing-Zhang cultural guide

Connect the Jing-Zhang Railway, Zhan Tianyou, Zhongguancun innovation, and AI self-restraint.

Physical labels, paper map, and a human guide independently deliver the full narrative

S02

Model red-team and failure drill

Reproduce harmful output, leakage, bias, and failed takeover before deployment.

Independent experts apply the same public checklist and publish findings beside model-based assurance

S05

Account-free learning and research explanation

Help the public understand corridor AI work, evidence, and applicability limits.

A staffed explanation, print reader, and device-free labels deliver the same learning goal

S08

Accessible service co-test

Let disabled users co-accept routes, wayfinding, assistance, and switching between service modes.

Physical wayfinding, staffed accompaniment, and a print checklist complete the same task

S11

Anonymous and non-personalized retail

Compare product discovery without a personal profile against staffed shopping.

A staff member and paper category index complete shopping at the same price and hours

S03

Camera-free edge perception benchmark

Test whether occupancy, obstacle, and queue assistance works without faces or identity.

Trained observers record the same non-identifying statistics during matched periods

S06

Health-service navigation without diagnosis

Explain public providers, opening hours, appointments, and accessible arrival options.

Phone, print directory, and trained staff complete the same search and appointment help

S09

Station-park transfer navigation

Combine walking, cycling, and public transport without claiming an engineering alignment.

Physical signs, a paper route map, and staffed help complete the same arrival task

S12

Face-free event flow

Use aggregate counts to operate events, shade, and dual-mode stormwater space.

Human patrol, physical zone signs, and public address deliver the same safety operation

The human route has an owner and a budget

RACI, OPEX, and independent evaluation enter the scale gate.

N	R01	SERVICE OWNER	Task, budget, and stop conditions
	R02	SITE OPERATOR	Staffing, maintenance, and takeover
	R03	INDEPENDENT EVALUATOR	Counterfactual gaps and red teaming
M	R04	COMMUNITY REPRESENTATIVE	Persona tasks and appeal oversight
	R05	DATA STEWARD	Clearance, minimization, and deletion
	R06	HUMAN REVIEWER	Final judgment, stop authority, and rest

PUBLIC BASE ROUTE / OPEX

FREE PUBLIC BASE ROUTE

Cross-subsidized by B2B testing, training, licensed events, and enterprise services; never by monetizing personal data.

SCALE GATE: BASELINE + SAFETY OWNER + OPEX

Every claim can be traced backward

source → geometry → metric → scenario → figure → risk

PROVISIONAL AREA

11.413

KM²

GREEN RATIO

15.29

%

NODE PUBLIC SPACE

0.89

%

NODES / PORTS / LANDMARKS

12 / 7 / 4



SOURCE

GEOMETRY

METRIC

SCENARIO

FIGURE

RISK

EVERY CLAIM CAN BE TRACED BACKWARD

Unknown metrics remain unknown; zero is never used as a substitute.

Unknown does not mean zero

Unknowns remain explicit for survey, clearance, baselining, and professional design.

UNKNOWN 01

BOUNDARY AND THREE KEY AREAS

Await registered official polygons

UNKNOWN 03

STATUTORY FAR / HEIGHT

Remain unknown; never infer

UNKNOWN 05

HUMAN-ROUTE OPEX

Budget gate at every phase

UNKNOWN 02

APPROX. 9 KM LINE

Requires measurement and definition clearance

UNKNOWN 04

HYDRAULICS AND POLLUTION

Require specialist models and sampling

UNKNOWN 06

CEI BASELINE

Target, not a current fact

WHAT DESERVES MEMORY MUST REMAIN CORRECTABLE

N

M

S



The monument records Agents, people, versions, licenses, and actual adoption state. Selection is not implementation; correction can update credit, version, and state.